

Research Question

- The 1875 Amalgamated Society of Railway Servants (ASRS) established branch networks across England
- Did proximity to union branches cause railway management to alter promotion practices?
- Two competing channels:
 1. **Management response:** Union threat slows promotion as discipline/deterrence
 2. **Composition:** ASRS collapse (1880–82) changed workforce composition via dismissals
- Identification: pre-1875 accident rates instrument for union branch location

Data Sources

Source	Coverage	N	Key Variables
ASRS Balance Sheets 1875	Nationwide	6,066 members; 139 branches	Name, occupation, join date, branch location
Railway Personnel Registers (Ancestry/TNA)	LNWR Birmingham; GWR Birmingham; GWR Ayles- bury	2,257 workers	Name, entry year, grade, promotion dates, station transfers
Accident Records (Board of Trade)	England, pre-1875	778 incidents	Location (lat/lon), cause, fatalities, in- juries

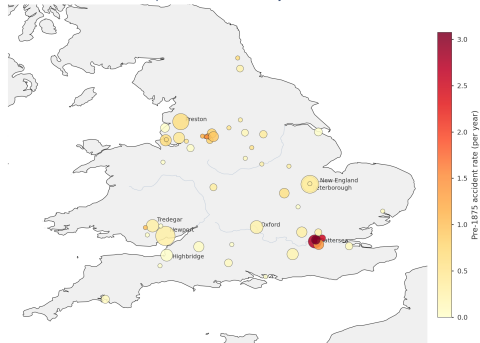
Proof of concept at LNWR Birmingham. Sheffield Midland register (1833–1888) download pending.

The Rise and Fall of the ASRS

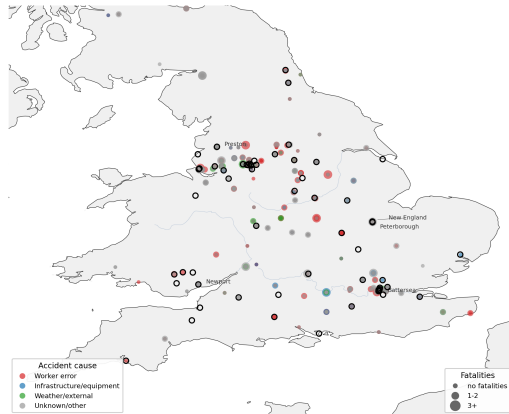
- **1871:** ASRS founded with support of Liberal MP Michael Bass
- **1872:** Peak membership of 17,247 — tiny fraction of 250,000 railway workers
- **1875:** Annual balance sheets record 139 branches nationwide; concerns over member fatalities drive organising
- **1879:** GWR dismisses petitioning members; companies increase working week to 66 hours
- **1880:** ASLEF founded — engine drivers and firemen split from ASRS
- **1882:** ASRS membership collapsed to 6,321
- **1913:** ASRS merges into National Union of Railwaymen

ASRS Branch Network and Pre-1875 Accident Geography

ASRS branches: membership and accident intensity



Pre-1875 accidents and ASRS branch locations



Accident Location Predicts Union Branch Location

$$\mathbf{1}\{\text{ASRS branch in RD}_r\} = \alpha + \beta \mathbf{1}\{\text{pre-1875 accident in RD}_r\} + X_r' \gamma + \delta_g + \varepsilon_r$$

	(1) Pooled	(2) County FE	(3) Division FE	(4) Line FE
Has accident	0.829*** (0.038)	0.830*** (0.039)	0.829*** (0.038)	0.832*** (0.047)
Observations	514	511	514	388
R^2	0.780	0.754	0.764	0.776
Fixed effects	None	County	Division	Railway line

Unit of observation is the 1851 Registration District with at least one rail station in England and Wales. Dependent variable equals one if the district has at least one ASRS branch. Key regressor equals one if the district has a pre-1875 railway accident. All columns control for $\log(\text{area}+1)$ and $\log(\text{stations}+1)$. HC1 robust standard errors in parentheses.

Register Description

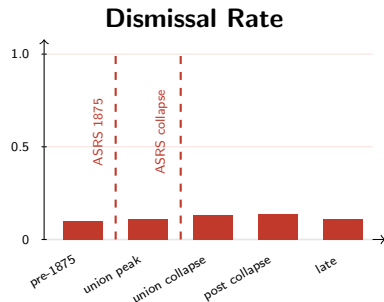
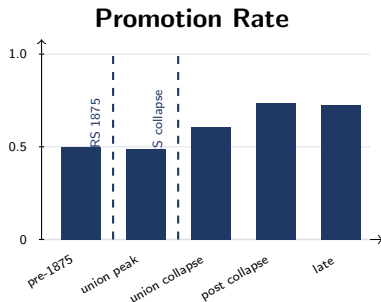
- Birmingham Central District Guards Register
- RAIL 410, Piece 0246 (TNA/Ancestry)
- Coverage: 1879–1924
- 423 workers with valid entry year
- Nearest ASRS branch: Dudley (13km)
- Format A: explicit grade progression dates recorded

Grade Progression

Grade	Description
Pit Goods Guard	Entry grade
Shunter	Intermediate
Goods Guard	Full promotion

Promotion = attaining Goods Guard or Shunter grade

Promotion and Dismissal Rates by Entry Cohort



Promotion rate: Share of workers who attained Goods Guard or Shunter grade within the observation window. LNWR Birmingham Guards register only ($n = 423$ after cleaning).

Dismissal rate: Share of workers whose remarks column contains keywords: *dismiss*, *discharg*, *resign*, *left service*, *removed*. Includes voluntary and involuntary exits; cannot distinguish between them.

Ruling Out Composition Effects

Full Sample

Period	Rate	N
pre-1875	0.500	10
union_peak	0.486	37
union_collapse	0.609	23
post_collapse	0.738	65
late	0.726	288

Excl. Engine Drivers & Firemen

Period	Rate	N
pre-1875	0.500	6
union_peak	0.472	36
union_collapse	0.722	18
post_collapse	0.789	57
late	0.774	226

The post-1880 recovery is not driven by engine drivers and firemen, the grades most directly affected by ASLEF's 1880 split.